

GIS-Based Flood Risk Assessment of Southwest Coastal Bangladesh Using a Multi-Criteria Weighted Overlay Approach

A GIS-based spatial assessment framework integrating topographic, hydrological, climatic, and land-use factors for flood susceptibility mapping.

Abstract

Flooding is one of the major environmental hazards affecting low-lying coastal regions, where topography, rainfall, river systems, and land surface characteristics influence flood susceptibility. Southwest coastal Bangladesh is highly vulnerable due to its flat terrain, extensive river networks, monsoon rainfall, and dynamic land-use conditions. This study develops a GIS-based flood risk assessment framework using a multi-criteria weighted overlay approach. Five major flood influencing factors, including elevation, slope, river proximity, rainfall, and land use/land cover (LULC), were analyzed using geospatial techniques. The individual risk layers were standardized and integrated using weighted overlay analysis with assigned weights of elevation (25%), river proximity (25%), slope (20%), LULC (15%), and rainfall (15%). The results demonstrate significant spatial variation in flood susceptibility across the study area. The final flood risk classification shows that Moderate Risk areas occupy the largest proportion (44.24%), followed by High Risk areas (32.24%) and Low Risk areas (19.50%). The developed GIS-based framework provides a spatial decision-support tool for identifying flood-prone zones and improving flood management, land-use planning, and disaster risk reduction strategies in coastal Bangladesh.

Keywords: Flood risk assessment; GIS; Weighted overlay analysis; Multi-criteria decision analysis; Coastal Bangladesh; Spatial modelling

1. Introduction

Flooding is one of the most frequent and damaging natural hazards worldwide, affecting human settlements, agriculture, infrastructure, and ecosystems. Coastal regions are particularly vulnerable because flood occurrence is controlled by the combined influence of terrain conditions, hydrological connectivity, rainfall variability, and land-use characteristics. Therefore, effective flood assessment requires an integrated approach that considers multiple environmental factors rather than relying on a single parameter.

Southwest coastal Bangladesh is highly susceptible to flooding due to its low-lying deltaic landscape, dense river networks, monsoon rainfall, and changing land-use patterns. The interaction between these factors creates complex flood dynamics, making spatial identification of vulnerable areas essential for effective planning and management.

Geographic Information System (GIS)-based multi-criteria analysis provides an effective approach for integrating diverse spatial datasets and evaluating flood susceptibility. By combining topographic, hydrological, climatic, and land surface information, GIS-based weighted overlay methods can generate comprehensive flood risk maps that support decision-making.

This study aims to develop a GIS-based flood risk assessment model for Southwest Coastal Bangladesh by integrating elevation, slope, river proximity, rainfall, and land use/land cover factors. The specific objectives are to (i) generate individual flood risk layers, (ii) integrate multiple factors through weighted overlay analysis, and (iii) produce a final flood risk map representing spatial variations of flood susceptibility.

2. Data and Methodology

2.1 Study Area and Dataset Description

The datasets used in this study were selected based on their ability to represent major physical and environmental controls influencing flood susceptibility. The Digital Elevation Model (DEM) was obtained from the Shuttle Radar Topography Mission (SRTM) dataset, which provides global elevation information for terrain analysis (Farr et al., 2007). Rainfall information was obtained from the Climate Hazards Infrared Precipitation with Station (CHIRPS) dataset, which provides long-term precipitation estimates suitable for hydrological applications (Funk et al., 2015). Land surface characteristics were represented using the ESA WorldCover 10 m land cover dataset (Zanaga et al., 2022), while river network information was obtained from global hydrographic datasets (Lehner & Grill, 2013). The characteristics, sources, and applications of the datasets used for flood risk assessment are summarized in Table 1.

Dataset	Source	Application
Digital Elevation Model	SRTM DEM	Elevation and slope analysis
Rainfall Dataset	CHIRPS	Rainfall risk assessment
Land Cover Dataset	ESA WorldCover	LULC risk analysis
River Network	Hydrological river dataset	River proximity analysis
Boundary Data	Administrative boundary data	Study area extraction

2.2 GIS Data Processing and Risk Layer Development

The analysis workflow consisted of data preparation, projection standardization, clipping, raster preprocessing, risk classification, raster alignment, and weighted overlay analysis.

3. Results and Discussion

3.1 Spatial Distribution of Flood Risk Components

The individual flood risk component maps were generated to evaluate the influence of different environmental factors controlling flood susceptibility. The spatial patterns of

elevation, slope, river proximity, rainfall, and land use/land cover demonstrate considerable variation across the study area. These factors represent the major physical and environmental conditions influencing flood generation and water accumulation processes.

As illustrated in Figure 1, each flood-controlling factor exhibits distinct spatial characteristics. The LULC risk map shows variations associated with different land surface conditions, while the river risk map highlights the influence of drainage networks on flood susceptibility. Similarly, rainfall and elevation patterns indicate areas where climatic and topographic conditions may increase flood potential. The combined interpretation of these individual factors provides the basis for developing the integrated flood risk model.

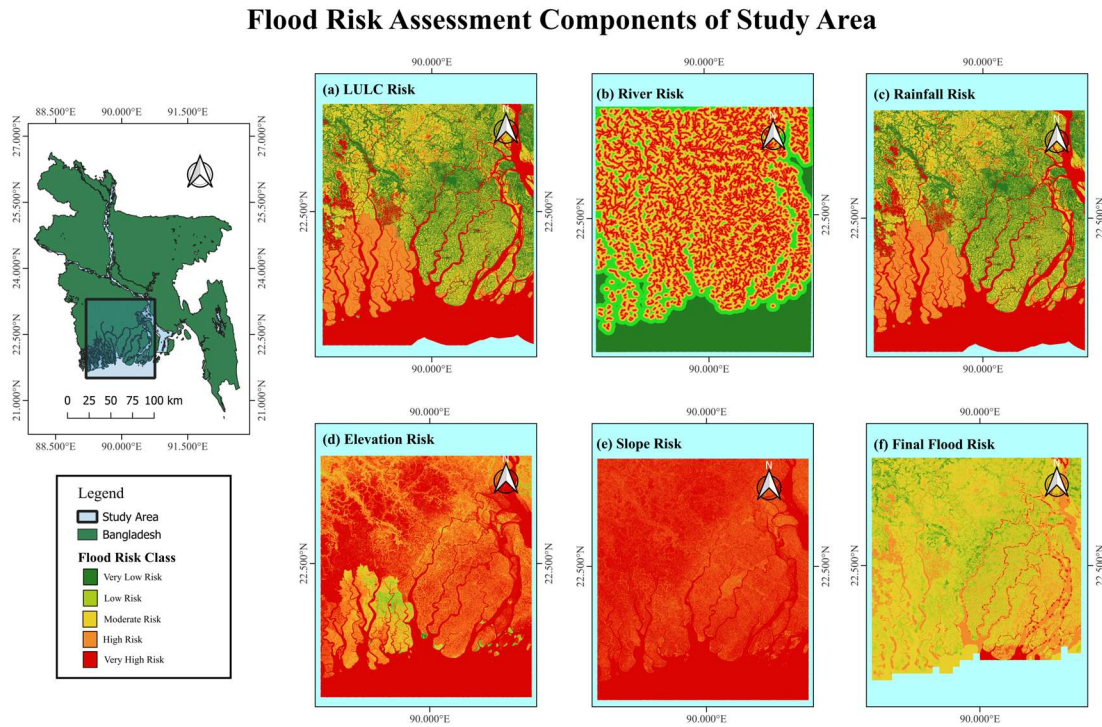


Figure 1. Spatial distribution of flood risk components: (a) LULC Risk, (b) River Proximity Risk, (c) Rainfall Risk, (d) Elevation Risk, (e) Slope Risk, and (f) Integrated Flood Risk Index.

3.2 Final Flood Risk Index Development

The individual risk layers were integrated using a weighted overlay approach to generate the final flood risk index. The weighted model considers the relative contribution of each factor based on its influence on flood susceptibility.

The final flood risk index was calculated using:

$$FRI = (Elevation \times 0.25) + (River \times 0.25) + (Slope \times 0.20) + (LULC \times 0.15) + (Rainfall \times 0.15)$$

The generated flood risk index represents the continuous spatial variation of flood susceptibility before classification into discrete risk categories.

3.3 Final Flood Risk Classification

The continuous flood risk index was further classified into different flood susceptibility categories to facilitate spatial interpretation and decision-making. The classified flood risk map represents the combined influence of topographic, hydrological, climatic, and land-use factors.

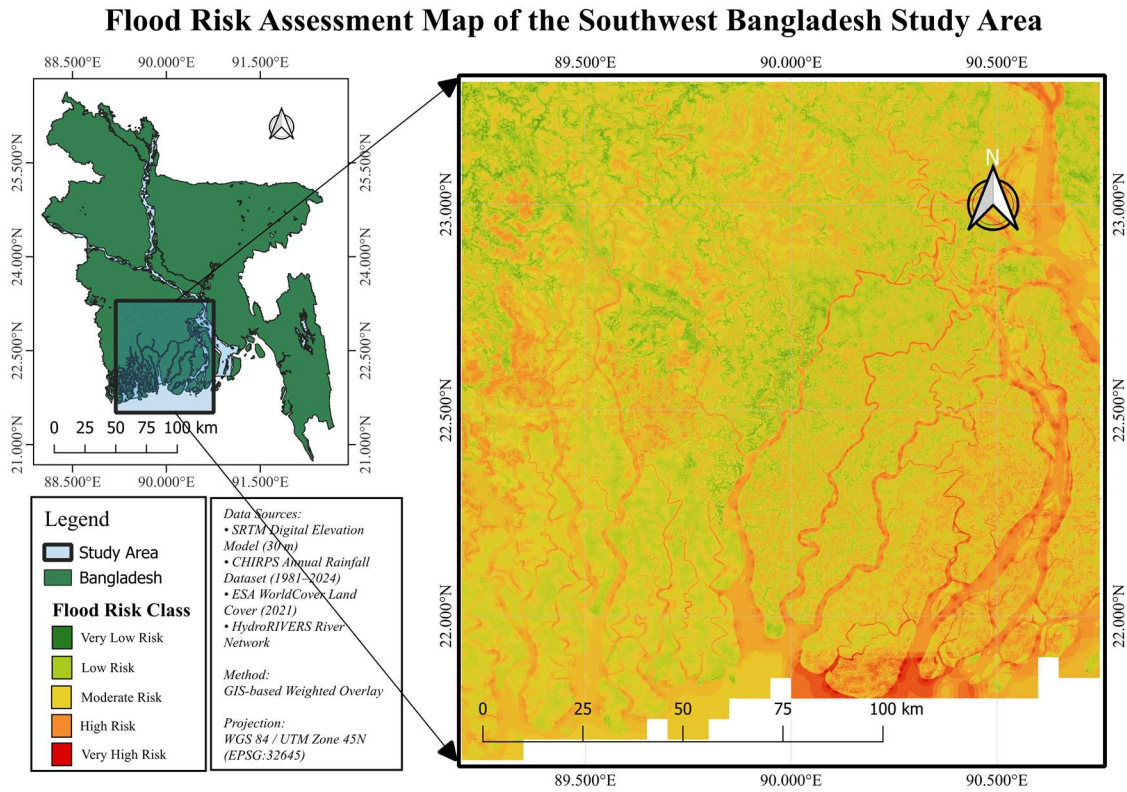


Figure 2. Final flood risk classification map of the Southwest Coastal Bangladesh study area.

As shown in Figure 2, the flood risk distribution varies spatially across the study area. Higher risk zones are mainly associated with areas where multiple flood-contributing factors interact, including low-lying terrain, proximity to river channels, higher rainfall influence, and less favorable surface conditions. In contrast, relatively lower risk zones occur in areas with comparatively favorable topographic and environmental characteristics.

3.4 Spatial Distribution of Flood Risk Classes

The final flood risk classification map was further analyzed to quantify the spatial distribution of different flood susceptibility zones within the study area. The classified flood risk raster was converted into area statistics to determine the contribution of each

risk category. The distribution of flood risk classes provides a quantitative understanding of the extent of areas exposed to different levels of flood susceptibility. The calculated area and percentage distribution of each flood risk category are presented in Table 2.

Table 2. Area and percentage distribution of different flood risk classes within the study area.

Flood Risk Class	Area (km ²)	Area (%)
Very Low Risk	179.78	0.66
Low Risk	5284.11	19.50
Moderate Risk	11991.23	44.24
High Risk	8739.44	32.24
Very High Risk	909.81	3.36

The flood risk distribution indicates that Moderate Risk areas represent the largest portion of the study area, followed by High Risk zones. Very High Risk areas are concentrated in locations influenced by multiple flood controlling factors.

4. Conclusion

This study successfully developed a GIS-based flood risk assessment framework for Southwest Coastal Bangladesh using a multi-criteria weighted overlay approach. The integration of elevation, slope, river proximity, rainfall, and LULC factors enabled the identification of spatial flood susceptibility patterns. The final flood risk map provides useful information for flood management, spatial planning, and future hazard assessment.

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